Internal Capital Frictions in Intermediaries and Exchange Rate Dynamics

Jean-Guillaume Magré[1] Gabor Pinter[2] Emil Siriwardane[3] Danny Walker[4]

[1]ECB and Oxford [2]BIS [3]HBS [4]Bank of England

Can the Convergence Risk and Internal Capital Frictions Explain FX Arbitrage Deviations?

Why CIP Arbitrage Is Actually Risky: The Convergence Risk

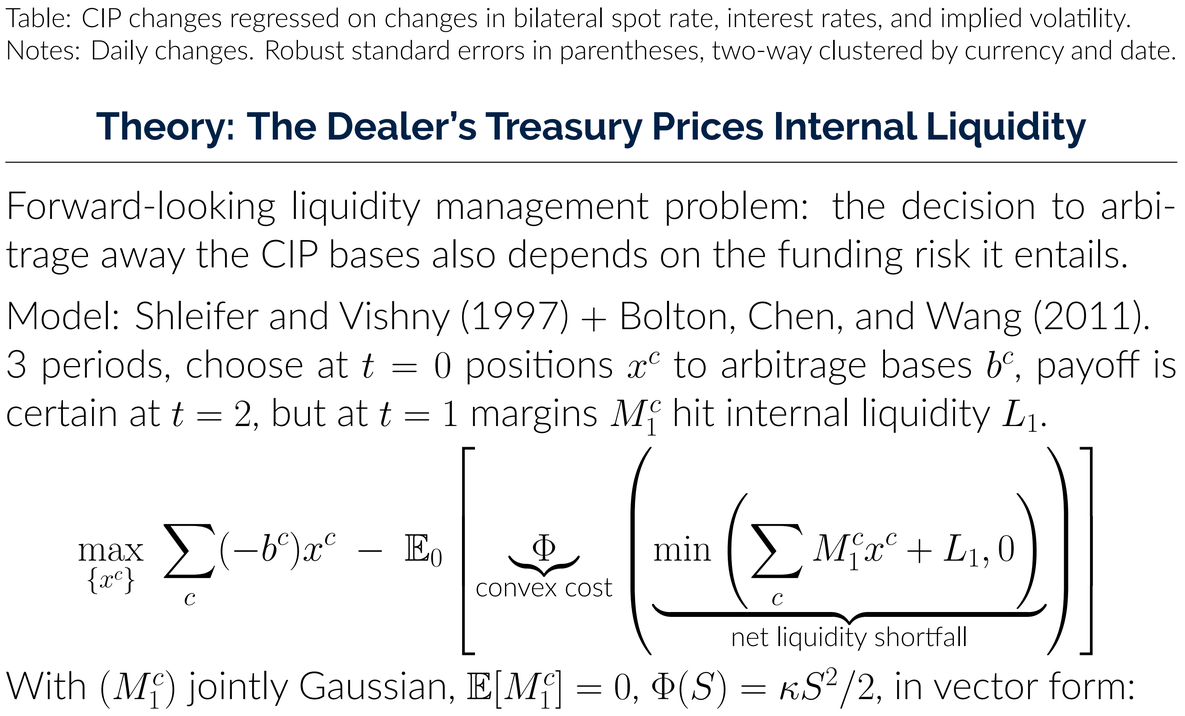
Significant changes in the mark-to-market values of outstanding derivative contracts of large dealers lead to sizable variation margins. These internal liquidity shocks prevent granular intermediaries from arbitraging CIP deviations away because of the marginal value of cash.

Channel: Variation Margins on FX Forwards Create Liquidity Risk

Covered Interest Rate Parity (CIP): $F_{0,T} = S_0 \frac{1+i_{0,T}-b_T}{1+i^*_{0,T}}$, with deviation b_T.

The puzzle: CIP bases $b_{c,T}$ are **not** equalized across currencies. Why?

Our hypothesis: variation margins M_t on the FX forward leg of the arbitrage create **convergence risk**, mark-to-market losses as prices move.



Back-of-the-envelope: 3m forward, $\sigma = 15\%$, get $MaR_{95\%} \simeq 12\% \times N$.

The currency implied vol is indeed associated with changes in CIP bases:

	$\Delta b_{c,t}$	$\Delta Spot_{c,t}$	$\Delta i_{c,t}$	$\Delta i_{USD,t}$	$\Delta IVol_{c,t}$
Coefficient		0.0231	-0.0417***	0.0905***	-0.0886***
Std. Error		(0.0901)	(0.0132)	(0.0156)	(0.0285)
Observations: 26,512			Number of currencies c: 9		R^2: 0.013

Table: CIP changes regressed on changes in bilateral spot rate, interest rates, and implied volatility.
Notes: Daily changes. Robust standard errors in parentheses, two-way clustered by currency and date.

Theory: The Dealer's Treasury Prices Internal Liquidity

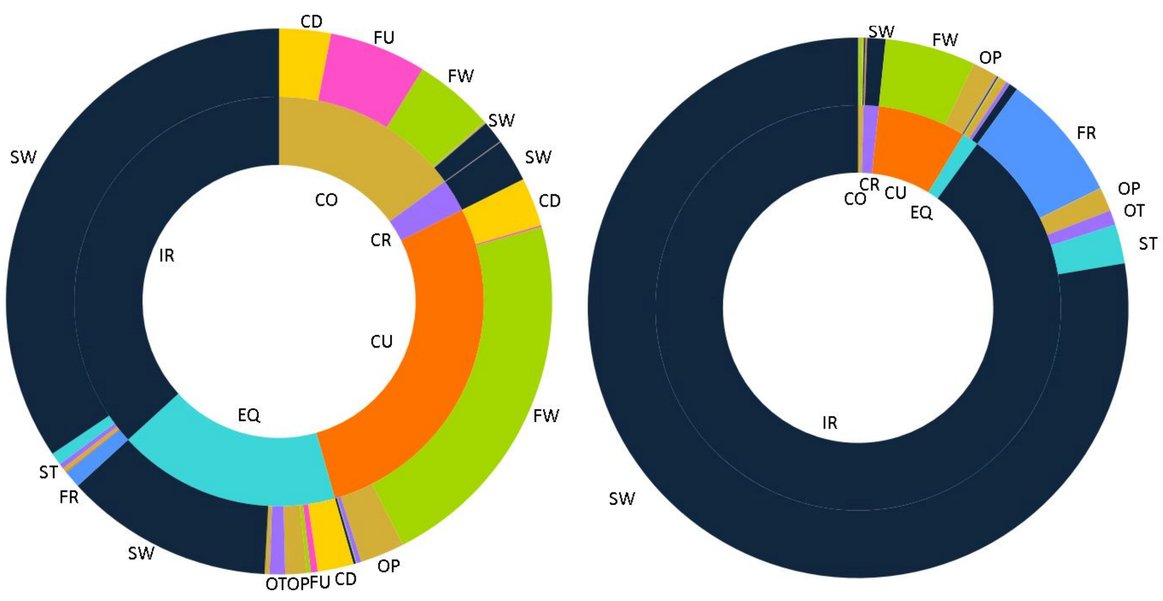
Forward-looking liquidity management problem: the decision to arbitrage away the CIP bases also depends on the funding risk it entails.

Model: Shleifer and Vishny (1997) + Bolton, Chen, and Wang (2011). 3 periods, choose at $t = 0$ positions x^c to arbitrage bases b^c, payoff is certain at $t = 2$, but at $t = 1$ margins M_1^c hit internal liquidity L_1.

$$\max_{\{x^c\}} \sum_c (-b^c)x^c \; - \; \mathbb{E}_0 \left[\underbrace{\Phi}_{\text{convex cost}} \left(\underbrace{\min\left(\sum_c M_1^c x^c + L_1, 0 \right)}_{\text{net liquidity shortfall}} \right) \right]$$

With (M_1^c) jointly Gaussian, $\mathbb{E}[M_1^c] = 0$, $\Phi(S) = \kappa S^2/2$, in vector form:

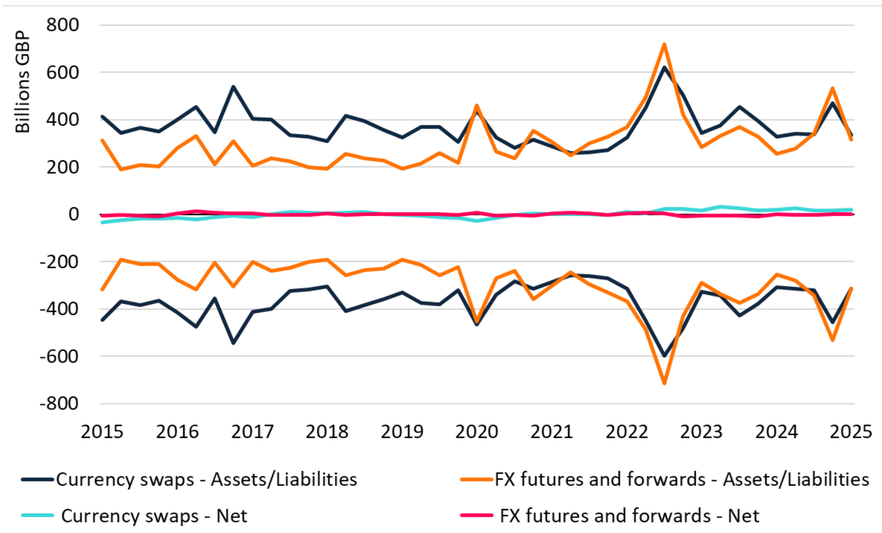
$$-b = \underbrace{\kappa}_{\text{cost of liquidity}} \underbrace{\Pr(x' M_1 + L_1 < 0)}_{\text{liquidity risk}} \underbrace{\Sigma}_{\text{variance-covariance}} \underbrace{x}_{\text{arbitrage position}}$$



FX forwards (UK-EMIR): 4 million transactions, USD 40 trillion notional.

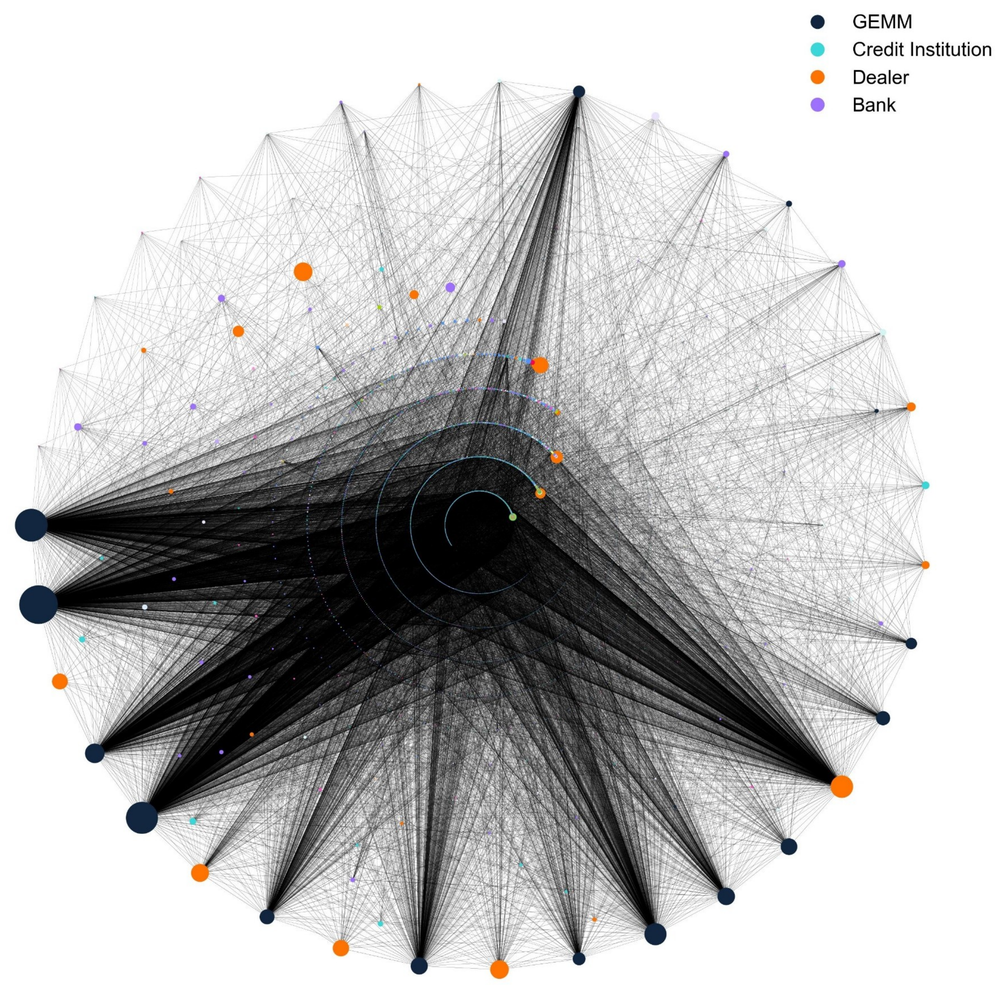
FX Forwards Market Structure: OTC and Highly Intermediated

Result 1: UK banks' FX forwards VMs $\simeq$ annual operating profits. Large!



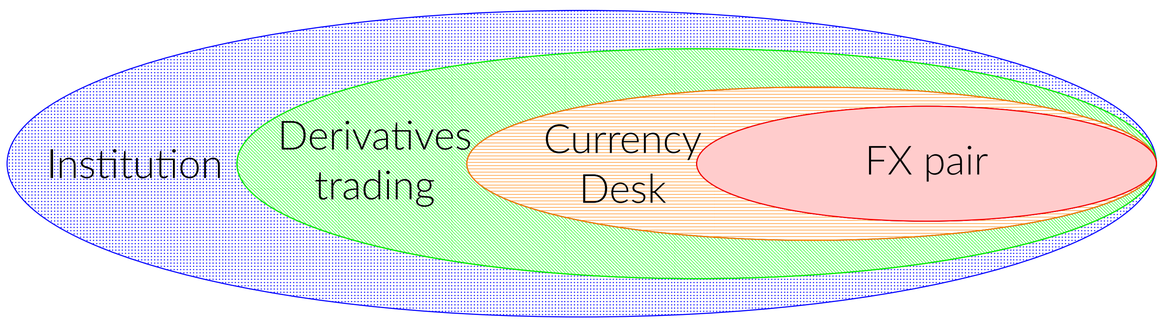
Result 2: Large, granular dealers intermediate the OTC, bilateral market.

➤ Dealers not substitutable: idiosyncratic shocks matter for CIP bases.



Result 3: Large dispersion in forward rates across dealers and clients.

Testing For Internal Frictions and Balance-Sheet Segmentation



What we find: evidence of balance-sheet segmentation, reject the hypothesis of a representative dealer sector, idiosyncratic shocks matter. Main specification: a USD 100 million shock widens CIP by 1bp (prov.).

Policy Implications: Variation Margins Are Not Liquidity-Neutral

Variation margins "only" move cash between counterparties, but this is not necessarily neutral for prices. Liquidity-preparedness frameworks should map margins to the business unit where liquidity binds.

CIP as a stress gauge. A wider basis can reflect tighter internal liquidity at a handful of dealers, not necessarily aggregate dollar scarcity.

Disclaimer: Any views expressed are solely those of the authors. These views should not be reported as representing the views of the ECB, the Eurosystem, the BIS, the Bank of England, or members of their committees.